

Project Report

CSE667 (DMMRS)

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1. Title of the proposed work (5 points):

Adversarial Contingency Auctions: Strategic Multi-Robot Task Allocation under Inconsistent Beliefs and Stochastic Infrastructure Collapse

2. Keywords (5 points)

Contingency-aware allocation, belief inconsistency, adversarial blockages, heterogeneous fleets

3. Impact of the work (*Maximum 1 page*) (10 points)

Autonomous multi-robot systems are being deployed into exactly the environments where our current theory breaks: contested disaster zones after earthquakes and floods, climate-stressed logistics corridors subject to rolling infrastructure failures, urban last-mile delivery networks under GPS spoofing, and hybrid civil-defence operations in post-conflict regions such as Ukraine and Gaza. In each of these 2024–2026 deployments, three pathologies appear simultaneously: robots cannot fully communicate (bandwidth is jammed, denied, or rationed), the environment mutates strategically (blockages are not merely stochastic but reflect adversarial or human-crowd responses to robot behaviour), and the fleet itself must be assembled on the fly from a heterogeneous pool of ground, aerial, and amphibious platforms with different costs and capabilities.

Existing frameworks address these pathologies in isolation. Contingency games (Peters et al., 2024) reason about future intent certainty but assume a fixed, fully communicating pair of agents. Stochastic-blockage task assignment (Wilde et al., 2022) handles recoverable environmental uncertainty but treats blockages as memoryless and non-strategic. Work on inconsistent-belief MR-BSP (Kundu et al., 2024) guarantees action consistency but presumes tasks are fixed and the environment static. Heterogeneous fleet design (Wilde et al., 2023) optimises composition offline under deterministic task loads. Distributed auction allocation (Bai et al., 2023) scales to hundreds of robots but collapses under belief divergence.

What is fundamentally missing is a unified framework in which (i) the fleet itself is a decision variable chosen under uncertainty about which tasks will even appear, (ii) robots bid on task bundles while holding provably inconsistent beliefs about the environment and about each other's beliefs, (iii) blockages are modelled as the output of a strategic adversary (congestion, jamming, hostile actors) rather than a Poisson process, and (iv) each robot plans a contingency tree over when its teammates' and adversary's intents will become clear. We call this regime **Adversarial Contingency Auctions (ACA)**.

The real-world payoff is direct. A 200-robot humanitarian logistics fleet in a flood zone must decide which drones and ground vehicles to deploy, which medical packages each

should bid on, and how to plan routes when half the team has stale maps, corridors are being blocked by displaced crowds reacting to robot presence, and the operator cannot afford to wait for consensus. ACA targets precisely this regime: massive scale, strategic uncertainty, and communication scarcity, where the cost of a wrong allocation is measured in lives rather than latency.

4. **Technicalities of the problem definition** (*Maximum 1 page*) (15 points)

Agents and fleet. Let $\mathcal{R} = \{R_1, \dots, R_I\}$ be a catalogue of robot types, each with capability set Ψ_i , cost b_i , battery β_i , and travel-time tensor D_i . A fleet $F \subseteq \mathcal{R}^N$ is drawn under a global budget B , so $\sum_{r \in F} b(r) \leq B$.

Tasks and environment. Tasks $\mathcal{T}_k = \{T_1, \dots, T_{\{n_k\}}\}$ arrive online; each $T_j = (v_j, t^r_j, t^d_j, \Phi_j)$ specifies a location, release/deadline window, and capability requirement. The environment is a graph $G = (V, E, \ell)$ in which a subset $E_{\text{risk}} \subset E$ is subject to blockages governed by a strategic adversary $\mathcal{A}$ with hidden type $\theta \in \Theta$, $|\Theta| = K$. Blockage evolution $\psi_t(\theta)$ depends on both θ and the fleet's joint action.

Beliefs, inconsistency, and branching. Each robot r maintains a private belief $b^r_k(\theta, x_k) = \mathbb{P}(\theta, x_k \mid \mathcal{H}^r_k)$ over the adversary type and world state, conditioned on its local history $\mathcal{H}^r_k$. Histories are generically inconsistent: $\mathcal{H}^r_k \neq \mathcal{H}^{r'}_k$, and communication is rationed by a per-epoch budget κ . Following contingency-game logic, each robot is parameterised by a branching time t^{br} at which it expects intent uncertainty about both $\mathcal{A}$ and its teammates to resolve.

Strategic interaction. At each planning epoch, robots play a contingency auction: each r computes, for every feasible bundle $\mathcal{G} \subseteq \mathcal{T}_k$ it could serve, a conditional trajectory tree $\tau^r_{\mathcal{G}}(\theta)$ branching at t^{br} , and submits a bid:

$$\beta^r(\mathcal{G}) = \mathbb{E}_{\theta \sim b^r_k} [c(\tau^r_{\mathcal{G}}(\theta))] + \lambda \cdot \mathcal{J}^r(\mathcal{G}, b^{\wedge\{-r\}}_k)$$

where $\mathcal{J}^r$ is a penalty capturing disagreement between r 's bid and what r infers its teammates would bid under their inferred beliefs $b^{\wedge\{-r\}}_k$, generalising action-consistency reasoning to the auction setting.

Objective. Jointly choose fleet F^* , allocation $\mathcal{A}^* : F \rightarrow 2^{\mathcal{T}_k}$, contingency policies $\{\tau^r\}$, and communication schedule $\mathcal{C}$ to solve:

$$\min_{\{F, \mathcal{A}, \{\tau^r\}, \mathcal{C}\}} \mathbb{E}_{\{\theta, \psi_t\}} [\sum_{r \in F} c(\tau^r_{\mathcal{A}(r)}(\theta)) + \mu \cdot \sum_j M \cdot \mathbb{1}[t^a_j > t^d_j]]$$

subject to: $\sum_r b(r) \leq B$, $|\mathcal{C}| \leq \kappa$

and capability, battery, and multi-robot action-consistency constraints.

Why it is fundamentally hard. It combines four sources of intractability: NP-hard fleet design, NP-hard task assignment with time windows, PSPACE-hard routing under adversarial blockages (a generalized Canadian Traveler Problem), and decentralized POMDP-style reasoning with inconsistent beliefs. These pieces are tightly coupled. The

fleet choice shapes the action space, which shapes the belief dynamics, which shapes when branching is worth paying for, which in turn changes the structure of each bid.

5. **Bibliography** (*Maximum ½ page*) (5 points)

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[2] N. Wilde and J. Alonso-Mora, "Online Multi-Robot Task Assignment with Stochastic Blockages," in Proc. 61st IEEE Conf. on Decision and Control (CDC), Cancún, Mexico, 2022, pp. 5259–5266.

[3] T. Kundu, M. Rafaeli, and V. Indelman, "Multi-Robot Communication-Aware Cooperative Belief Space Planning with Inconsistent Beliefs: An Action-Consistent Approach," in Proc. IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS), Abu Dhabi, UAE, 2024, pp. 13221–13228.

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[5] X. Bai, A. Fielbaum, M. Kronmüller, L. Knoedler, and J. Alonso-Mora, "Group-Based Distributed Auction Algorithms for Multi-Robot Task Assignment," IEEE Trans. on Automation Science and Engineering, vol. 20, no. 2, pp. 1292–1303, Apr. 2023.